JOÃO VICTOR WALCACER GIANI

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LinkedIn | GitHub

EDUCATION

Federal University of Tocantins, Brazil *March 2022 - August 2026*
Bachelor's Degree in Computer Science
Machine Learning, Software Engineering, Artificial Intelligence, Cloud Computing, Distributed Systems, Algorithms, Databases, Operating Systems, Compilers, Graph Theory, Statistics, Data Structures, Algorithms.

WORK EXPERIENCE

Junior Software Engineer *April 2026 - Present*
Vetta Digital *Remote, Brazil*

- Reduced frontend server CPU and memory consumption by 40%, based on comparative runtime metrics analysis, by isolating responsibilities into a dedicated backend API service implemented with **Node.js and PostgreSQL**. Structured the solution using **Clean Architecture, DDD, and Modular Monolith**, centralizing data processing and access in the backend and using **Swagger** for API documentation and standardization, improving application efficiency and scalability.
- Reduced the loading time of a screen processing more than 90,000 records by 80%, from approximately 5s to 1s. Implemented **Limit/Offset pagination in PostgreSQL** to reduce network payload size and per-request processing, using **TanStack Query (React Query)** for asynchronous data fetching, **Infinite Scroll**, **dynamic Query Keys**, **caching and invalidation**, **predictive Prefetching**, and a **Stale-While-Revalidate** strategy, minimizing latency and optimizing incremental data loading.
- Optimized the processing of large volumes of data on the frontend, reducing the complexity of certain operations from **$O(n^2)$ to $O(n)$**. Replaced repetitive linear searches with **Hash Map** data structures, pre-indexing data to enable direct access to the required information and reduce preprocessing time before rendering.

Software Engineer Intern *April 2024 - April 2026*
Vetta Digital *Remote, Brazil*

- Developed **reusable components** for different application contexts, reducing logic duplication and increasing consistency across screens. Used **React and TypeScript**, with strong typing, composition, and well-defined component APIs, as well as **Styled Components** for styling. Also contributed to the internal component library, using **Storybook and JSDoc** for documentation and usage examples, facilitating adoption and maintenance by other developers.
- Improved the user experience across the application's main screens by implementing **Skeleton Loading, Empty States, and loading and error states** in asynchronous flows. Using **React and TypeScript**, developed visual states that preserved the interface structure during loading and provided contextual feedback when no data was available or an error occurred, reducing perceived latency and making navigation more predictable.

Automation Engineer Intern *April 2023 - April 2024*
Brain Analytics *Remote, Brazil*

- Reduced the execution time of an operational task from approximately **30 to 5 minutes** by automating the collection and preparation of data from multiple platforms and sources. Developed the solution using **Python, Pandas, and Selenium**, performing data extraction, integration, cleaning, and preprocessing before making the data ready for employees, eliminating manual and repetitive steps.

- Developed an automation for credit availability analysis, reducing the time required to process and respond to requests and supporting decision-making. Implemented the workflow using **Python and FastAPI** to integrate with internal APIs, **PostgreSQL** for data storage, and a Machine Learning pipeline with **Scikit-learn and Random Forest**. Trained the model with data provided by the bank to generate **reliability scores**, using **Joblib** for model serialization and deployment within the prediction workflow.

Software Engineer *January 2025 - April 2026*
Direct Clients *Remote, Brazil*

- Developed an automated **document verification** system for a client in the automotive sector, integrating CRM APIs and combining **Machine Learning, Computer Vision, and LLMs** in a validation pipeline. Used **Python and FastAPI**, training **Azure** image recognition models through iterative training, evaluation, and feedback cycles, and integrated LLMs via **Google AI Studio** for text information extraction and validation. Implemented **structured JSON outputs** for programmatic response validation and resilience mechanisms such as **retries with exponential backoff and jitter, fallbacks, and circuit breaker** to handle external service failures.
- Developed an asynchronous integration between a **CRM webhook** and a messaging application, decoupling event reception from processing and enabling horizontal scalability. Implemented the solution with **Python, FastAPI, BullMQ, and Redis**, using **queues and workers** for asynchronous processing, as well as **idempotency, structured logs, and correlation IDs** mechanisms to ensure reliable processing and end-to-end traceability.

PROJECTS

Amamenta Palmas — Human Milk Bank Management System
Academic Project — GitHub

- Led the development of a management system for a **hospital Human Milk Bank**, developed in partnership with the university and hospital to digitalize processes such as donor registration, visit scheduling, milk collection, pasteurization, inventory management, and milk distribution. Acted as **Tech Lead and responsible for the technical direction and product evolution**, designing the architecture and coordinating the development of the solution.
- Developed the web application with **React, TypeScript, and Vite**, using **React Query** for client-side data management and synchronization, **Axios** for API consumption, and **Styled Components and Storybook** for componentization, styling, and documentation of the component library.
- Designed and developed the backend as a **Modular Monolith** using **Node.js, TypeScript, Fastify, Drizzle ORM, and PostgreSQL**, applying **Clean Architecture** to separate business rules from infrastructure. Implemented contracts and validations with **Zod**, authentication with **JWT**, and API documentation with **Swagger**, structuring the system to facilitate its evolution and eventual module extraction.

Chronic Disease Prediction with Social Determinants of Health
Bachelor's Thesis — GitHub

- Developed a modular predictive architecture to investigate the influence of **Social Determinants of Health (SDH)** on hypertension prediction, using microdata from the **National Health Survey (PNS 2019)** and applying the **Knowledge Discovery in Databases (KDD)** process for data preparation, analysis, and modeling.
- Implemented and evaluated supervised **Machine Learning** models, comparing **Logistic Regression with LASSO regularization** and **Random Forest** across biomedical, social, and integrated scenarios. The integrated models achieved an **AUC close to 0.80 and recall of approximately 75%**, maintaining discriminative capability even with the inclusion of sociodemographic variables.
- Analyzed the influence and fairness of predictions using **Odds Ratio, Feature Importance, and fairness metrics**, identifying performance differences across education-level groups and quanti-

fying the influence of socioeconomic factors on the results. Developed the analysis pipeline in **Python**, using **JupyterLab** and Machine Learning and data processing libraries.

RAG Chatbot with Vector Database
Personal Project — GitHub

- Developed a **Retrieval-Augmented Generation (RAG)** POC for generating contextual responses from custom documents, implementing the complete pipeline for **document upload, chunking, embeddings, vector search, and response generation**. Used **Python, Django REST Framework, LangChain, and Google Gemini** for LLM and embedding orchestration, with **PostgreSQL + pgvector and FAISS** for vector storage and retrieval.
- Implemented asynchronous document indexing using **Celery and Redis**, as well as real-time chat with **Django Channels, WebSockets, and Daphne**. Developed the frontend with **React, TypeScript, and Vite**, using **Axios** for API communication, and containerized the application with **Docker and Docker Compose**.

TECHNICAL SKILLS:

Frontend: React, TypeScript, JavaScript, React Query, Vite, Axios, Styled Components, Storybook

Backend & APIs: Node.js, Fastify, Python, FastAPI, Django, Django REST Framework, REST APIs, WebSockets, Webhooks, Drizzle ORM, Zod, JWT

Data & AI: PostgreSQL, SQL, Redis, Pandas, Scikit-learn, LLMs, RAG, LangChain, FAISS, pgvector, Computer Vision, Selenium

Architecture & Engineering: Clean Architecture, DDD, Modular Monolith, Multi-tenant

Cloud, DevOps & Tools: Microsoft Azure, Docker, Docker Compose, Swagger/OpenAPI, BullMQ, Celery, Git

PERSONAL TRAITS

- Strong communication skills, developed through practical experience in professional environments, academic presentations, and participation in seminars and technology events.
- Ability to learn new concepts and adapt to different technologies, with a strong interest in continuous learning.
- Collaborative profile, with the ability to work effectively in teams and take on leadership responsibilities when necessary.

LANGUAGES:

Portuguese: Native

English: Upper-Intermediate to Advanced (B2/C1)